



KARN HR ACADEMY

Production Possibility Curve (PPC)

Simple Notes

Study Notes | For academic use only

Production Possibility Curve (PPC) — Simple Notes

1. What is a PPC?

A country has **limited resources** (land, labour, capital). It can use them to make different goods, so it must **choose**. The PPC is a curve that shows **all the possible combinations of two goods** a country can produce with its given resources. It is also called the **production possibility frontier** or **transformation curve** (one good is 'transformed' into another by shifting resources).

Assumptions: (i) only two goods, (ii) resources are fixed, (iii) all resources are fully and efficiently used, (iv) technology does not change.

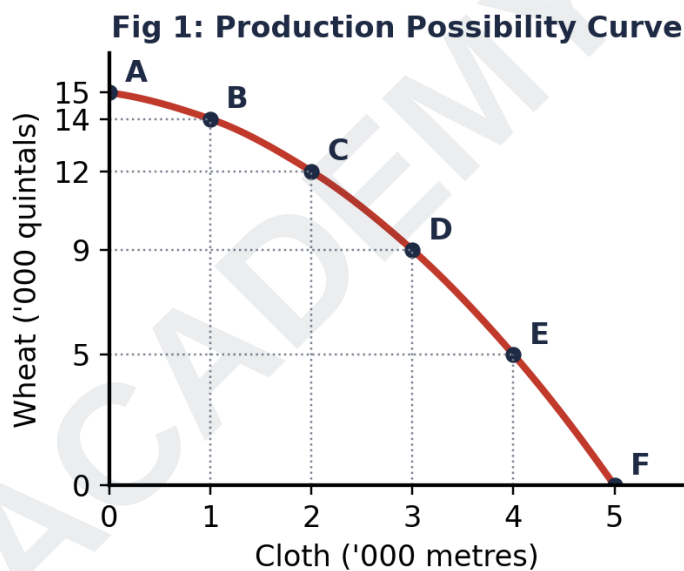
2. Simple example: Wheat vs Cloth

Suppose a country makes only **wheat** and **cloth**. If all resources go to wheat → 15 (thousand quintals). If all go to cloth → 5 (thousand metres). In between are other choices:

Choice	Cloth	Wheat	Wheat given up
A	0	15	—
B	1	14	1
C	2	12	2
D	3	9	3
E	4	5	4
F	5	0	5

Cloth in '000 metres, wheat in '000 quintals.

Plotting these points gives the PPC (Fig 1). **More cloth means less wheat.**



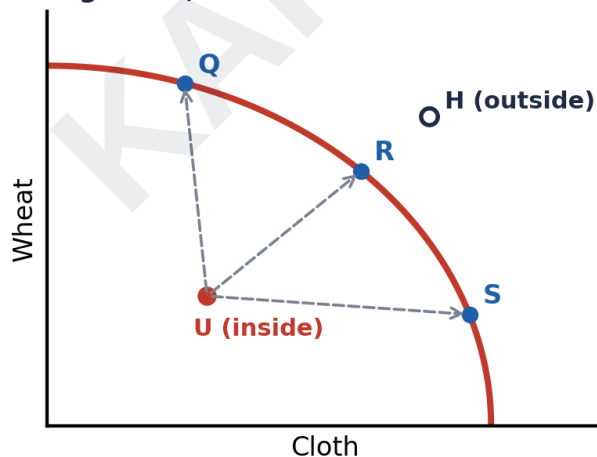
3. Why is the curve bowed outward (concave)?

Opportunity cost = how much of one good we give up to get one more unit of the other. In the table, each extra unit of cloth costs 1, then 2, 3, 4, 5 units of wheat — the cost **keeps rising**. This is the **law of increasing opportunity cost**.

Reason — specificity of resources: not all resources are equally good for both goods. Fertile farm land is great for wheat but not for cloth. When we keep shifting to cloth, we must use resources that are better at wheat, so we lose more and more wheat.

4. Points on, inside and outside the curve

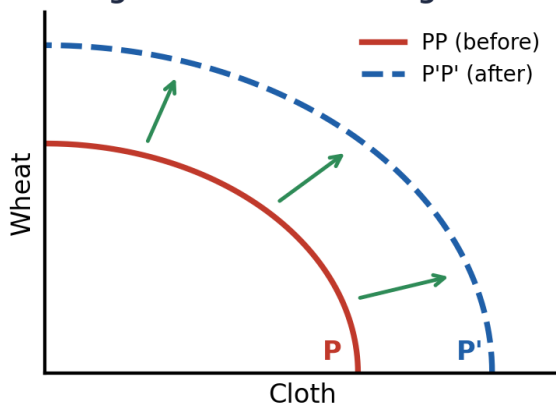
Fig 2: On, inside & outside the curve



- **On the curve (Q, R, S):** all resources are fully and efficiently used. Maximum output.
- **Inside the curve (U):** some resources are idle or wasted → **unemployment / inefficiency**. By using idle resources the economy can move from U to Q, R or S and get more of one or both goods.
- **Outside the curve (H):** **not possible** with present resources and technology.

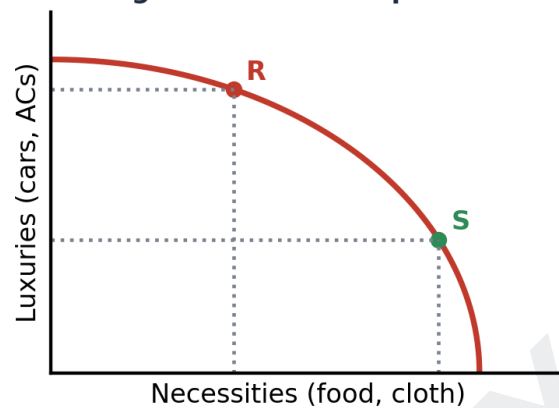
Example: if factories are shut during a slowdown, the country is at U. Keynes suggested raising demand (e.g., public works) to bring it back onto the curve.

Fig 3: Outward shift = growth



5. Economic growth shifts the PPC outward. If resources increase (more land, labour, capital) or **technology improves**, the country can make **more of both goods**. The curve moves from PP to P'P'.
Example: better seeds and new machines → more wheat and and more cloth.

Fig 4: For whom to produce



6. For whom to produce? Put necessities on one axis, luxuries on the other.
 • At **R**: many luxuries, few necessities → output favours the **rich**, unequal distribution.
 • At **S**: more necessities → more **equal** distribution.

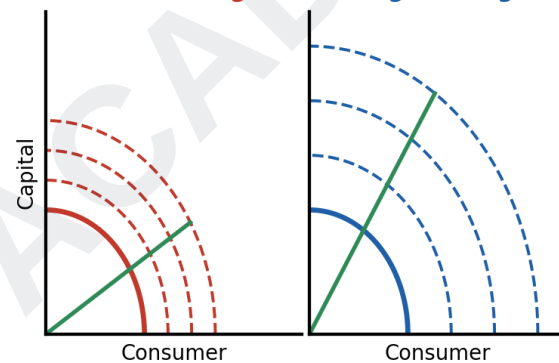
Fig 5: Consumer vs capital goods



7. Capital goods vs consumer goods. Moving from A to B means making more machines (capital) but fewer consumer goods **today**.
 More machines raise future capacity → more consumer goods **tomorrow**: “less jam today for more jam tomorrow.”

Fig 6: Growth paths

Low saving **High saving**



8. Rate of growth. Each dashed curve is the PPC in a later year; the green line is the growth path.
 • **Low saving** (less capital formation) → curve moves out slowly.
 • **High saving** (more capital formation) → curve moves out faster, so more goods in the future.

9. PPC and the basic economic problems (in one line each)

- **Scarcity & choice:** resources are limited, so more of one good means less of the other.
- **What to produce:** which point on the curve the country chooses (in a free market, decided by consumer demand).
- **How to produce:** using efficient methods keeps the economy on the curve, not inside it.
- **For whom to produce:** the mix of necessities and luxuries shows who gets the output (Fig 4).
- **Growth:** more resources, capital and technology push the whole curve outward (Figs 3, 5, 6).

Quick recap: PPC = all possible combinations of two goods · bowed out because of increasing opportunity cost · on = efficient, inside = unemployment, outside = impossible · outward shift = economic growth.